

1. The Autonomous "Greenwashing" & ESG Compliance Auditor

- **The Real-World Business Problem:** Global regulatory bodies are penalizing corporations millions of dollars for "greenwashing"—claiming their supply chains are sustainable when they are actually causing environmental damage. Compliance officers must manually audit thousands of supply chain invoices, transport logs, and factory images to verify these claims.
- **How the Tech Blends:**
 - **Multimodal AI:** Processes drone or satellite imagery of factory locations (Vision) alongside unstructured text from global shipping manifests.
 - **Hybrid RAG Pipeline:** Ingests the latest corporate ESG (Environmental, Social, Governance) laws and international carbon tax code regulations.
 - **LLM + Multi-Agent Orchestration:**
 - *Agent 1 (The Visual Forensic Agent):* Scans the factory images to detect unauthorized deforestation, unmapped waste runoff, or lack of solar infrastructure.
 - *Agent 2 (The Data Miner):* Cleans and processes the raw shipping telemetry and invoices using **Pandas** and **NumPy**.
 - *Agent 3 (The LLM Legal Reasoner):* Takes the outputs from the first two agents, queries the RAG database, and acts as the central brain. It evaluates the alignment between the company's public sustainability claims and the actual data, outputting a mathematically explainable "Deception Risk Score" and a fully cited compliance report.

2. Generative Film Continuity & "Script-to-Screen" Director

- **The Real-World Business Problem:** In film production, maintaining continuity across months of shooting is an expensive nightmare. A character wearing a watch on their left wrist in one scene might have it on their right wrist in the next day's shoot, costing thousands of dollars in post-production visual effects to fix.
- **How the Tech Blends:**
 - **Multimodal AI:** Ingests live video clips from the day's raw video footage (rushes) and reads character placement, background lighting, and actor positioning.
 - **GraphRAG Pipeline:** A complex Knowledge Graph that indexes the entire official movie script, character lore books, and physical prop placement registries.

- **LLM + Multi-Agent Orchestration:**
 - *Agent 1 (The Visual Continuity Agent):* Watches the recorded scene frames, tracking actor pose estimation, costume details, and prop geometry.
 - *Agent 2 (The Script Supervisor Agent):* Listens to the dialogue audio tracks to check if the actors accurately delivered the script lines without breaking character lore.
 - *Agent 3 (The LLM Director):* Serves as the central reasoning node. It compares the visual/audio outputs against the GraphRAG script using long-context LLM reasoning to catch discrepancies (e.g., "The script dictates it is raining outside, but the video shows bright sunshine through the window") and instantly logs a redline ticket for the production team.

3. Generative Patent Infringement Defiler & Claim Constructor

- **The Real-World Business Problem:** Tech companies spend millions of dollars hiring patent attorneys to draft new patent claims or evaluate if a new competitor's product infringes upon their existing intellectual property. The text in patents is notoriously complex, highly dense, and legally specific.
- **How the Tech Blends:**
 - **Multimodal AI:** Parses structural engineering blueprints, system architecture flowcharts, and technical CAD drawings submitted for the patent.
 - **Hybrid RAG Pipeline:** Connected to the massive, live global patent registry database, utilizing BM25 search for exact legal clauses and dense vector search for technological concepts.
 - **LLM + Multi-Agent Orchestration:**
 - *Agent 1 (The Blueprint Extractor):* Translates the visual layout and structural diagrams into detailed text descriptions of the mechanical or digital architecture.
 - *Agent 2 (The Patent Searcher):* Queries the global RAG database to pull similar existing patents.
 - *Agent 3 (The LLM Claim Evaluator):* A high-powered LLM reasoning agent that conducts a semantic cross-examination of the new product vs. the retrieved patents. It breaks down the claim language line-by-line, identifies potential infringement vulnerabilities, and

autonomously rewrites the legal claims to ensure the new patent is bulletproof before filing.

4. Smart Grid Dynamic Pricing & Asset Health Command Center

- **The Real-World Business Problem:** Electrical grids face massive load spikes during extreme heatwaves or local events, causing transformer failures and localized blackouts. Operators need a system that doesn't just display telemetry charts, but actually reasons about operational risks in real time.
- **How the Tech Blends:**
 - **Multimodal AI:** Ingests thermal imagery of electric substations (Vision) alongside massive tabular time-series streams of grid voltage, phase fluctuations, and ambient weather temperatures.
 - **RAG Pipeline:** Vectorizes heavy industrial electrical equipment OEM manuals, historical failure logs, and grid safety operating procedures.
 - **LLM + Multi-Agent Orchestration:**
 - *Agent 1 (The Anomaly Tracker):* Flags abnormal acoustic or thermal spikes in localized substations.
 - *Agent 2 (The Data Engineer):* Computes rolling energy anomalies across different zones using data engineering pipelines.
 - *Agent 3 (The LLM Operations Commander):* Acts as the core cognitive layer. It reads the telemetry context and the flagged visual risks, consults the RAG manuals, and applies LLM reasoning to predict structural failure windows. It then dynamically drafts optimization recommendations for the grid control room to reroute power automatically.

Team Execution Strategy

Because you prefer to tackle the core coding logic yourself, your group can partition these architectures cleanly:

1. **Member 1** can own the multimodal ingestion framework (OpenCV, MediaPipe, or vision transformers).
2. **Member 2** handles the complex data pipelines, vector database optimization, and cloud storage layout.
3. **Member 3** constructs the core LLM prompt boundaries, **Streamlit** user experience, and LangGraph orchestrations.